

Michail Mitsakis

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Education

- MSc** [UNIVERSITY], Materials Science [START – END]
- Thesis: optimisation of electrodeposition parameters for hydrogen evolution reaction (HER) catalysts.
 - Specialisation: electrochemistry and green hydrogen.
- BSc** [UNIVERSITY], Physics [START – END]

Experience

- [CONSULTANCY NAME], [ROLE TITLE] Greece [START] – present
- Client-facing delivery on deep-tech and EU-funded programmes in hydrogen technologies, manufacturing data spaces, and shipyard digitalisation.
 - Proposal development, technical scoping, and meeting preparation with industrial and research partners.

Projects

- catalyst-kg-agent** Jan 2026
- Knowledge-graph-grounded, cost-aware multi-agent system for materials-discovery campaigns.
- Decides between a knowledge-graph lookup, an MLIP surrogate query (MACE), and an expensive simulated experiment under an explicit budget.
 - Materials Project knowledge graph of 189 HER/OER-relevant materials; zero-shot MACE reaches 0.114 eV/atom MAE on non-oxides.
 - CGCNN baseline evaluated with composition-disjoint cross-validation to control polymorph leakage.
 - Stack: PyTorch Geometric, MACE, BoTorch/Ax, pydantic-ai, Ollama, NetworkX, MLflow.
- notion-second-brain** Jan 2026
- Fully local retrieval-augmented agent over a personal Notion workspace.
- Hybrid dense + BM25 retrieval with reciprocal-rank fusion and cross-encoder reranking.
 - Persistent memory, OCR ingestion of PDFs and images, and an anchored-rubric evaluation harness.
 - Stack: pydantic-ai, Ollama, Qdrant, Streamlit, Docker.
- tender-feed** Jan 2026
- Scheduled monitor for public tenders and funding calls with weekly email digests.
- Configurable per-source scraping with seen-item state and bilingual (English/Greek) keyword prioritisation, run weekly on GitHub Actions.

Skills

Electrochemistry

Electrocatalysis, hydrogen evolution reaction, electrodeposition

Machine learning for materials

Bayesian optimisation, graph neural networks, machine-learned interatomic potentials (MACE), Materials Project

Software

Python, PyTorch Geometric, BoTorch/Ax, pydantic-ai, LLM agents, RAG, MLflow, Docker, GitHub Actions

Project delivery

EU-funded project proposals, technical scoping, consortium work